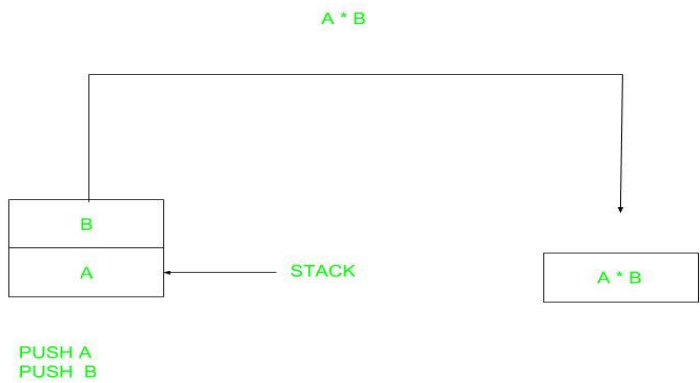


Instruction format:

On the basis of number of address instruction are classified as:

Note that we will use $X = (A+B)*(C+D)$ expression to showcase the procedure.

1. Zero Address Instructions –



A stack based computer do not use address field in instruction.To evaluate a expression first it is converted to revere Polish Notation i.e. Post fix Notation.

Expression: $X = (A+B)*(C+D)$

Postfixed : $X = AB+CD+*$

TOP means top of stack

$M[X]$ is any memory location

PUSH	A	TOP = A
PUSH	B	TOP = B
ADD		TOP = A+B
PUSH	C	TOP = C
PUSH	D	TOP = D
ADD		TOP = C+D
MUL		TOP = (C+D)*(A+B)

POP	X	$M[X] = \text{TOP}$
-----	---	---------------------

2. One Address Instructions –

This use a implied ACCUMULATOR register for data manipulation. One operand is in accumulator and other is in register or memory location. Implied means that the CPU already know that one operand is in accumulator so there is no need to specify it.

opcode	operand/address of operand	mode
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Expression: $X = (A+B)*(C+D)$

AC is accumulator

M[] is any memory location

M[T] is temporary location

LOAD	A	$AC = M[A]$
------	---	-------------

ADD	B	$AC = AC + M[B]$
-----	---	------------------

STORE	T	$M[T] = AC$
-------	---	-------------

LOAD	C	$AC = M[C]$
------	---	-------------

ADD	D	$AC = AC + M[D]$
-----	---	------------------

MUL	T	$AC = AC * M[T]$
-----	---	------------------

STORE	X	$M[X] = AC$
-------	---	-------------

3. Two Address Instructions –

This is common in commercial computers. Here two address can be specified in the instruction. Unlike earlier in one address instruction the result was stored in accumulator here result can be stored at different location rather than just accumulator, but require more number of bit to represent address.

opcode	Destination address	Source address	mode
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Here destination address can also contain operand.

Expression: $X = (A+B)*(C+D)$

R1, R2 are registers

M[] is any memory location

MOV	R1, A	$R1 = M[A]$
ADD	R1, B	$R1 = R1 + M[B]$
MOV	R2, C	$R2 = C$
ADD	R2, D	$R2 = R2 + D$
MUL	R1, R2	$R1 = R1 * R2$
MOV	X, R1	$M[X] = R1$

4. Three Address Instructions –

This has three address field to specify a register or a memory location. Program created are much short in size but number of bits per instruction increase. These instructions make creation of program much easier but it does not mean that program will run much faster because now instruction only contain more information but each micro operation (changing content of register, loading address in address bus etc.) will be performed in one cycle only.

opcode	Destination address	Source address	Source address	mode
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Expression: $X = (A+B)*(C+D)$

R1, R2 are registers

M[] is any memory location

ADD	R1, A, B	$R1 = M[A] + M[B]$
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ADD	R2, C, D	$R2 = M[C] + M[D]$
MUL	X, R1, R2	$M[X] = R1 * R2$